

ROC Curve and AUC Score

Machine Learning Notes

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1 Introduction

ROC stands for

Receiver Operating Characteristic Curve.

ROC Curve is used to evaluate the performance of a Binary Classification model.

Instead of checking only Accuracy, ROC Curve tells us how well the classifier separates positive and negative classes.

Examples:

- Cancer Detection
- Spam Email Detection
- Fraud Detection
- Diabetes Prediction
- Loan Approval

2 Why Accuracy is Not Enough?

Suppose we have

- 990 Healthy Patients
- 10 Cancer Patients

If the model predicts everyone as Healthy,
Accuracy becomes

$$\frac{990}{1000} = 99\%$$

Accuracy looks excellent.

But

The model failed to detect every cancer patient.

Hence Accuracy alone is misleading.

ROC Curve solves this problem.

3 Confusion Matrix

For Binary Classification,

	<i>PredictedPositive</i>	<i>PredictedNegative</i>
<i>ActualPositive</i>	<i>TP</i>	<i>FN</i>
<i>ActualNegative</i>	<i>FP</i>	<i>TN</i>

Where

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

4 True Positive Rate (TPR)

Also called

Recall

or

Sensitivity

Formula

$$TPR = \frac{TP}{TP + FN}$$

Example

Suppose

TP = 90

FN = 10

Then

$$TPR = \frac{90}{90 + 10} = 0.90$$

Meaning

The model correctly detects

90%

of positive samples.

5 False Positive Rate (FPR)

Formula

$$FPR = \frac{FP}{FP + TN}$$

Example

FP = 20

TN = 180

$$FPR = \frac{20}{20 + 180} = 0.10$$

Meaning

10%

of negative samples were incorrectly classified as positive.

6 ROC Curve

ROC Curve plots

- X-axis = False Positive Rate (FPR)
- Y-axis = True Positive Rate (TPR)

Different thresholds produce different TPR and FPR values.
 Joining these points creates the ROC Curve.
 The closer the curve is to the top-left corner,
 the better the classifier.

7 Example

Suppose model predicts probabilities

Actual Class	Probability
1	0.95
1	0.88
0	0.76
1	0.70
0	0.55
0	0.32
0	0.12

If threshold changes
 from
 0.9
 to
 0.5
 more observations become Positive.
 Hence
 TPR increases
 and
 FPR also increases.
 This generates ROC Curve.

8 Area Under Curve (AUC)

AUC means
 Area Under ROC Curve.
 It summarizes ROC Curve into one number.

$$0 \leq AUC \leq 1$$

Interpretation

AUC	Interpretation
1.00	Perfect Model
0.95	Excellent
0.90	Very Good
0.80	Good
0.70	Fair
0.60	Poor
0.50	Random Guess

9 Python Example

```
1
2 from sklearn.datasets import load_breast_cancer
3
4 from sklearn.model_selection import train_test_split
5
6 from sklearn.linear_model import LogisticRegression
7
8 from sklearn.metrics import roc_curve, roc_auc_score
9
10 import matplotlib.pyplot as plt
11
12 data=load_breast_cancer()
13
14 X=data.data
15
16 y=data.target
17
18 X_train,X_test,y_train,y_test=train_test_split(
19 X,y,test_size=0.2,random_state=42)
20
21 model=LogisticRegression(max_iter=10000)
22
23 model.fit(X_train,y_train)
24
25 prob=model.predict_proba(X_test)[: ,1]
26
27 fpr,tpr,threshold=roc_curve(y_test,prob)
28
29 auc=roc_auc_score(y_test,prob)
30
31 print("AUC =",auc)
32
33 plt.plot(fpr,tpr,label="ROC Curve")
34
35 plt.plot([0,1],[0,1], 'r--')
36
37 plt.xlabel("False Positive Rate")
38
39 plt.ylabel("True Positive Rate")
40
41 plt.legend()
42
43 plt.show()
```

10 Explanation of Code

Step 1

Load Dataset

```
load_breast_cancer()
```

Loads Breast Cancer dataset.

Step 2

Split Dataset

Training

80%

Testing

20%

Step 3

Train Logistic Regression

```
model.fit()
```

Learns relationship between features and target.

Step 4

Predict Probability

```
predict_proba()
```

Returns

Probability of each class.

We use

`[:, 1]`

because ROC needs

Probability of Positive Class.

Step 5

Calculate ROC

```
roc_curve()
```

Returns

- False Positive Rate
- True Positive Rate
- Thresholds

Step 6

Calculate AUC

```
roc_auc_score()
```

Calculates area under ROC Curve.

11 Advantages

- Independent of Threshold
- Better than Accuracy
- Useful for Imbalanced Dataset
- Compares Multiple Models
- Easy to Interpret

12 Disadvantages

- Only for Classification
- Harder to Interpret than Accuracy
- Doesn't Tell Probability Calibration
- PR Curve may be better for highly imbalanced data

13 Interview Questions

1. What is ROC Curve?
2. What is AUC?
3. Why Accuracy is misleading?
4. What is TPR?
5. What is FPR?
6. Why do we use `predict_proba()`?
7. Difference between ROC and Precision-Recall Curve?
8. What does $AUC = 0.5$ mean?
9. Can ROC be used for Regression?
10. Which model has better ROC?

14 Conclusion

ROC Curve is one of the most important evaluation metrics for Binary Classification.

Instead of relying only on Accuracy, it measures the model's ability to distinguish between positive and negative classes across different thresholds.

A higher AUC value indicates a better classifier.